

Riccardo Rettore

Robotics & Physical AI engineer passionate about humanoids, embodied intelligence, and robot autonomy

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EXPERIENCE

• Co-Founder & CTO @ Resit

Dec 2025 - Present

Smart Seating for healthier employees - resitchair.com

Padova - San Francisco

- Leading the development of learning-based posture classification models, transforming multi-modal sensor data into real-time user state estimation and ergonomic insights.
- Driving system architecture and integration across embedded sensing, ML inference, and user-facing applications, enabling closed-loop intelligence from physical interaction to real-time feedback.

• CDEI Group @ University of Catalunya

Feb 2025 – Aug 2025

[Prof. A. Gracia](#) (UPC), [Prof. G. Susto](#) (UniPd)

Barcelona, Spain

- Developed reinforcement learning-based navigation and control frameworks for autonomous robots using custom **Gymnasium** environments, implementing **PPO** and **SAC** in discrete and continuous action spaces with **Python** and **PyTorch**.
- Integrated subsurface perception from Ground Penetrating Radar with surface sensing into decision-making pipelines, validating navigation policies in **ROS2**-based simulations with **Gazebo** and visualizing results in **RViz**.

• Freelancer for Startups

Jan 2020 - Jan 2024

Full-Stack Developer

Padova, Italy

- Collaborated with startups to build cross-platform applications using **Flutter** and **Google technologies**.
- Built both front-end interfaces and back-end systems to deliver seamless, user-friendly solutions.

EDUCATION

• MSc in Control Systems Engineering

Oct 2023 - Oct 2025

Università degli Studi di Padova, Final Grade: 110/110

Padova, Italy

- Final Thesis: [Leveraging Ground Penetrating Radar with Deep Reinforcement Learning for Subsurface-Informed Navigation in Autonomous Agricultural Robotics](#).
- Relevant Coursework: Reinforcement Learning, Advanced Control Theory, Controls for Robotics Systems, Deep Learning, Machine Learning, Computer Vision, Autonomous Robotics.
- Erasmus Exchange Program @ Universitat Politècnica de Catalunya, Barcelona, Spain: focused on Reinforcement Learning and Robotics, Physical AI and Autonomous Systems.

• BSc in Mechatronics Engineering

Oct 2020 - July 2023

Università degli Studi di Padova, Final Grade: 109/110

Padova, Italy

- Final Thesis: [Integration between ROS2 and Unity: performance analysis](#).
- Relevant Coursework: Applied Mechanics, Analog Electronics, Electrical Machines and Drives, Instrumentation and Measurement, PLCs and Industrial Networks.

ACADEMIC EXPERIENCE

• Aurora: Object Classification with Oriented Boosting

July 2024 - Nov 2024

Neural Networks and Deep Learning project

Padova, Italy

- Built a 3D object classification pipeline with multi-task CNNs and oriented boosting, including voxelization, augmentation, labeling, and HDF5 preprocessing using **PyTorch**, **NumPy**, and **OpenCV**.
- Optimized GPU training and evaluated on ModelNet10/40, executing 3D data processing, CNN optimization, and computer vision pipelines.

• Advanced Control Systems for Double-Wheel Robot

April 2024 - June 2024

Control Engineering Laboratory Project

Padova, Italy

- Designed nominal and robust LQR controllers for a self-balancing two-wheeled robot, modeling dynamics via Lagrange formalism, linearizing around equilibrium, and implementing sensor fusion with complementary filtering in **MATLAB/Simulink**.
- Performed simulation and on-hardware testing of designed controllers, analyzing system stability, state-feedback, observer performance, and system robustness.